EXP20

```
import pandas as pd

import numpy as np

from sklearn.linear_model import LinearRegression


# Sales dataset

data = {

   "Month": [1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12],

   "Sales": [12000, 13500, 15000, 16000, 17500, 19000,

             20500, 22000, 23500, 25000, 27000, 29000]

}


df = pd.DataFrame(data)


# Input and output

X = df[["Month"]]

y = df["Sales"]


# Train model

model = LinearRegression()

model.fit(X, y)


# Predict future sales

future_months = np.array([[13], [14], [15], [16], [17]])


predictions = model.predict(future_months)


# Display results

for month, sales in zip(future_months.flatten(), predictions):

   print("Month", month, "Predicted Sales: ₹", round(sales, 2))
```

OUTPUT

Month 13 Predicted Sales: ₹ 29795.45

Month 14 Predicted Sales: ₹ 31302.45

Month 15 Predicted Sales: ₹ 32809.44

Month 16 Predicted Sales: ₹ 34316.43

Month 17 Predicted Sales: ₹ 35823.43s